Question	Answer	Marks	AO	Guidance	
7	[Perimeter =] $2r + r\theta = 10$	M1	1.1		
	$\theta = \dfrac{10-2r}{r}$	B1	3.1a	OR $r = \dfrac{10}{2+\theta}$	Expression for one of r, θ in terms of the other
	$A = \dfrac{1}{2}r^2\theta$ so $A = \dfrac{r(10-2r)}{2} = 5r - r^2$	M1	3.1a	OR $A = \dfrac{100\theta}{2(2+\theta)^2} = \dfrac{50\theta}{(2+\theta)^2}$	Area in terms of either *their* r or *their* θ. Need not expand brackets.
	$A = 2.5^2 - (2.5-r)^2$	M1	3.1a	Completing the square	
	This has a max when $2.5 - r = 0$	B1	2.4	Convincing explanation that there is a max	
	Max = 6.25 [cm^2]	A1	2.2a		
	Alternative method 1				
	$\dfrac{dA}{dr} = 5 - 2r$	M1			
	$\dfrac{d^2A}{dr^2} = -2$ so max	B1			
	$\dfrac{dA}{dr} = 0 \Rightarrow r = 2.5$; Max = 6.25 [cm^2]	A1			
	Alternative method 2				
	$\dfrac{dA}{d\theta} = \dfrac{50(2+\theta)^2 - 100\theta(2+\theta)}{(2+\theta)^4} = \dfrac{100-50\theta}{(2+\theta)^3}$	M1		Reasonable attempt at quotient rule	
	$\dfrac{dA}{d\theta} = 0 \Rightarrow \theta = 2$; A = 6.25 [cm^2]	A1		For information	
	$\theta = 1.5 \Rightarrow \dfrac{dA}{d\theta} > 0$, $\theta = 2.5 \Rightarrow \dfrac{dA}{d\theta} < 0$ so max	B1		$\dfrac{d^2A}{d\theta^2} = -\dfrac{200}{256} = -0.78125$ at $\theta = 2$	
		[6]			